B3 Problems: Geometry, proof, Pythagoras, similarity and trigonometry

← Module overview · Solutions

Draw a sketch for every problem. Give a reason for every angle you find. Round lengths to 2 decimals and angles to 1 decimal of a degree, unless the problem asks for an exact answer. Calculator in degree mode.

Practice

1. Lines ℓ and m are parallel. A line t crosses ℓ at P and m at Q. At P, the angle above ℓ and to the right of t is 68°.

(a) Find the other three angles at P, with reasons.
(b) Find all four angles at Q, with reasons (F-angles, Z-angles).
(c) Which two angles at P and Q are co-interior? Check that they add up to 180°.

2. (a) The angles of a triangle are x, $2x$ and $3x$. Find them. What kind of triangle is it?

(b) A triangle has angles 47° and 68°. Find the exterior angle at the third vertex in two ways.
(c) In triangle ABC, the exterior angle at C is 130° and $\angle A$ is 25° more than $\angle B$. Find $\angle A$ and $\angle B$.

3. (a) Find the angle sum of a 9-gon.

(b) Find each angle of a regular 12-gon.
(c) A regular polygon has angles of 156°. How many sides does it have?
(d) *(Challenge)* Draw a five-pointed star (a pentagram) by joining every second vertex of any convex pentagon. Prove that the five angles at the points of the star add up to 180°. (Hint: exterior angle theorem.)

4. For each pair of triangles ABC and DEF, decide whether they must be congruent. If yes, name the congruence condition (English and Dutch). If no, explain why.

(a) $AB = DE$, $BC = EF$, $\angle B = \angle E$.
(b) $\angle A = \angle D$, $\angle B = \angle E$, $AB = DE$.
(c) $AB = DE$, $AC = DF$, $\angle B = \angle E$, where AC is the shorter of the two given sides.
(d) $\angle C = \angle F = 90°$, $AB = DE$, $AC = DF$.
(e) $\angle A = \angle D$, $\angle B = \angle E$, $\angle C = \angle F$.
(f) $\angle A = \angle D$, $\angle B = \angle E$, $BC = EF$.

5. (a) In triangle ABC, $AB = AC$ and M is the midpoint of BC. Prove, in two-column form with *gegeven* and *te bewijzen*, that $AM \perp BC$ and that AM bisects $\angle A$.

(b) AB is a diameter of a circle with centre O, and C is on the circle with $\angle CAB = 34°$. Find $\angle ABC$ and $\angle BOC$. If $AB = 10$ and $AC = 6$, find BC.

6. In triangle ABC the angle at C is 90°.

(a) $AC = 7$, $BC = 24$. Find AB.
(b) $AB = 13$, $BC = 5$. Find AC.
(c) $AC = 5$, $BC = 8$. Find AB exactly and rounded.

7. Decide whether each triangle is right, acute or obtuse. Justify with the converse of Pythagoras.

(a) Sides 20, 21, 29. (b) Sides 8, 15, 16. (c) Sides 6, 7, 10. (d) Sides $\sqrt{5}$, $\sqrt{11}$, 4.

8. Triangle ABC has sides $AB = 6$, $BC = 8$, $AC = 10$. Triangle DEF has sides $DE = 9$, $EF = 12$, $DF = 15$.

(a) Show that the triangles are similar and give the scale factor from ABC to DEF.
(b) Show that $\triangle ABC$ is right-angled and compute its area.

(c) Compute the area of $\triangle DEF$ in two ways.

9. (A-figure) In triangle ABC, D lies on AB and E on AC with $DE \parallel BC$. $AD = 4$, $DB = 6$, $DE = 5$, $AE = 3$.

(a) Explain why $\triangle ADE \sim \triangle ABC$.
(b) Find BC, AC and EC.

10. (X-figure) Segments AD and BC cross at S, and $AB \parallel CD$. $AS = 6$, $SD = 9$, $AB = 8$, $BS = 5$.

(a) Explain why $\triangle SAB \sim \triangle SDC$.
(b) Find CD and SC.

11. Triangle ABC has a right angle at C, $AC = 12$ and $\angle A = 35°$. Find BC and AB.

12. A right triangle has legs 5 and 9. Find both acute angles and the hypotenuse.

13. Without a calculator, give exact answers.

(a) $\sin^2 45° + \cos^2 60°$
(b) $\tan 30° \cdot \tan 60°$
(c) $\sin 60° \cos 30° - \cos 60° \sin 30°$
(d) α is acute and $\sin \alpha = \frac{3}{5}$. Find $\cos \alpha$ and $\tan \alpha$.

14. A scale model of a church is made at scale $1 : 200$.

(a) The roof of the model has area 50 cm². What is the roof area of the real church, in m²?
(b) The model has volume 1500 cm³. What is the volume of the real church, in m³?

School-exam level

15. (Ladder) A ladder 6.50 m long leans against a vertical wall. Its foot stands on horizontal ground, 1.80 m from the wall.

(a) How high up the wall does the ladder reach?
(b) Find the angle between the ladder and the ground.
(c) The foot slips outwards until it is 2.30 m from the wall. How far does the top slide down?
(d) A safety rule says the angle between ladder and ground must be between 70° and 76°. Is the ladder safe before the slip? After the slip?

16. (Ramp) A wheelchair ramp must overcome a height difference of 0.45 m. Suppose the rule is that the slope may be at most $1 : 12$ (at most 1 m up per 12 m horizontally).

(a) What is the smallest possible horizontal length of the ramp?
(b) How long is the sloping surface of that ramp?
(c) Find the slope angle (*hellingshoek*) and the slope as a percentage.

17. (Tower) From point A on flat ground, the top of a church tower is seen at an angle of elevation of 32°. After walking 40 m straight towards the tower, to point B, the angle of elevation is 51°. Ignore eye height.

(a) Show that the height h of the tower satisfies $\dfrac{h}{\tan 32°} - \dfrac{h}{\tan 51°} = 40$.
(b) Find h, and the distance from B to the foot of the tower.

18. (Box) A closed box has the shape of a cuboid $ABCD.EFGH$ with $AB = 60$ cm, $BC = 40$ cm and $AE = 30$ cm.

(a) Compute the diagonal AC of the bottom face.
(b) Compute the space diagonal AG.
(c) Can a thin rod of 75 cm lie flat on the bottom of the box? Can it fit in the box at all?
(d) Find the angle between AG and the bottom face.

19. (Pyramid) A square pyramid $T.ABCD$ has base side 6 and height 4. The apex T is straight above the centre M of the base. N is the midpoint of BC.

(a) Compute TN and the edge TA (exact and rounded).
(b) Find the angle between edge TA and the base.
(c) Find the angle between face TBC and the base.
(d) Compute the total surface area.
(e) Describe the front view (*vooraanzicht*, looking horizontally, straight at face TAB) and the top view (*bovenaanzicht*) with all measurements. Which length in the front view is *not* the edge length TA?

20. (Sight lines) Lisa stands on a platform; her eyes are 4.50 m above the ground. A wall 3.00 m high stands 6.00 m in front of her (horizontal distance from her eyes to the wall). The ground is flat. Treat the wall as having no thickness.

(a) How deep is the blind spot (*dode hoek*) on the ground behind the wall?
(b) Tom, 1.80 m tall, stands 5.00 m behind the wall. Can Lisa see any part of him? How much?
(c) How high would Lisa's eyes need to be for the blind spot to be only 4.00 m deep?

21. (Shadow) A lamppost 4.80 m high lights up a flat street at night. Ella (1.60 m) stands 3.00 m from the foot of the lamppost.

(a) How long is her shadow?
(b) She walks away from the lamppost. Show that her shadow is always half as long as her distance to the lamppost.

22. (Exact values) A regular hexagon has side 4.

(a) Explain why it can be cut into six equilateral triangles.
(b) Find the exact height of one of these triangles and the exact area of the hexagon. Then round the area.
(c) Find the exact distance between two opposite sides of the hexagon.

23. (Two right triangles) In triangle ABC, $\angle A = 40°$, $\angle B = 65°$, and the altitude from C meets AB at D, with $CD = 5$.

(a) Why may you not use SOS-CAS-TOA directly in triangle ABC?
(b) Compute AD, DB and AB.
(c) Compute AC, BC and the area of triangle ABC.

Challenge

24. (Median to the hypotenuse) Triangle ABC has a right angle at C, and M is the midpoint of AB. Prove that $MC = \frac{1}{2}AB$. Then prove the converse: if the median CM equals half of AB, then $\angle C = 90°$.

25. (Angle bisector theorem) In triangle ABC, the bisector of angle A meets BC at D.

(a) Prove that $\dfrac{BD}{DC} = \dfrac{AB}{AC}$. (Hint: draw the line through C parallel to AD; let it meet line AB at E. Look for an isosceles triangle and an A-figure.)
(b) If $AB = 6$, $AC = 9$ and $BC = 10$, find BD and DC.

26. (Circumcentre) Prove that the three perpendicular bisectors of the sides of any triangle pass through one point O, and that the circle with centre O through A also passes through B and C. (You may assume that two of the perpendicular bisectors intersect.)

27. (Pythagorean triples)

(a) Find all right triangles with whole-number sides in which one leg is 12. (Hint: $c^2 - b^2 = (c - b)(c + b)$.)

(b) Prove that for whole numbers $m > n \geq 1$ the numbers $m^2 - n^2$, $2mn$, $m^2 + n^2$ form a Pythagorean triple. Which m, n give the triples from (a), if any?

28. (Cube) A cube has edge a. Two space diagonals cross at the centre of the cube. Find the angle between them (the acute angle and the obtuse angle). Explain why your answer does not depend on a.

29. (Altitude on the hypotenuse) In triangle ABC with a right angle at C, the altitude CD has length h, and D divides AB into $AD = q$ and $DB = p$.

(a) Prove that $h^2 = pq$.
(b) Prove that $ab = ch$, where $a = BC$, $b = AC$, $c = AB$.
(c) If $p = 9$ and $q = 4$, find h, a and b exactly and check (b).

30. (Incircle of a right triangle) A circle inside triangle ABC touches all three sides (the incircle, *ingeschreven cirkel*), with centre I and radius r. The angle at C is 90°.

(a) Prove that the two tangent segments from a vertex to the circle are equal. (Use RHS: the radius to a point of tangency is perpendicular to the side.)
(b) Prove that $r = \dfrac{a + b - c}{2}$.
(c) Find r for the triangles 3-4-5, 5-12-13 and 8-15-17. What do you notice?